Chapter 01: Introduction to Discrete Data in R

> "Essentially, all models are wrong, but some are useful."
>
> — George E.P. Box
>
> "You can have data without information, but you cannot have information without data."
>
> — Daniel Keys Moran

What are models?

Models propose a simplified representation of the reality, which is useful to develop a common ground for describing, analyzing and understanding complex phenomena.

Model building requires three things:

1. Raw materials.
2. Tools.
3. Technical expertise (hopefully!).

This is true whether the model is physical (for instance a sculpture), conceptual (a mental map), or statistical/mathematical (the gravity model or a regression model).

In the case of a sculpture, the raw materials can be marble, wood, or clay; the tools chisels, mallet, and spatula; and the technique the mastery of the sculptor when working with the tools and the materials. Anyone can try sculpture, and most people can create sculptures. These kind of models are evaluated by their aesthetic value, not necessarily their usefulness. But if the sculpture is poorly balanced and falls and breaks, then its value is limited by its structural integrity - the skill of the sculptor matters even if only in this sense.

In the case of a mental map, the raw materials are ideas, the tools are a drawing surface and tools for writing, or maybe an app, and the technical expertise is the ability of the modeler to organize ideas in a useful way. There are useful conceptual models, and conceptual models that are less useful. Figure @ref(fig:fig-conceptual-models) shows two example of conceptual models. The first example on the left, while displaying creative graphic features, is rather difficult to make sense of. The second one, on the right, welcomes a much more intuitive interpretation, with simplified graphic features.

In the case of mathematical/statistical models, the raw materials are data; the tools are descriptive statistics and statistical plots, and various forms of regression analysis; and the technical expertise is the ability of the modeler to select tools that are appropriate to the data, and to bring the data to "speak": in other words, to extract information from the data. As Moran said in the aphorism quoted at the top of this section, you can have data without information, but no information without data. Technical mastery is the degree to which a modeler can obtain information from data that is useful, accurate, and precise, to the extent that the raw materials permit.

Before moving on to the technical skills required for modeling, it is important to understand the raw materials and the tools. The objective of this note is to introduce some important concepts concerning data and data manipulation, and some useful tools, with an emphasis on discrete data unique features.

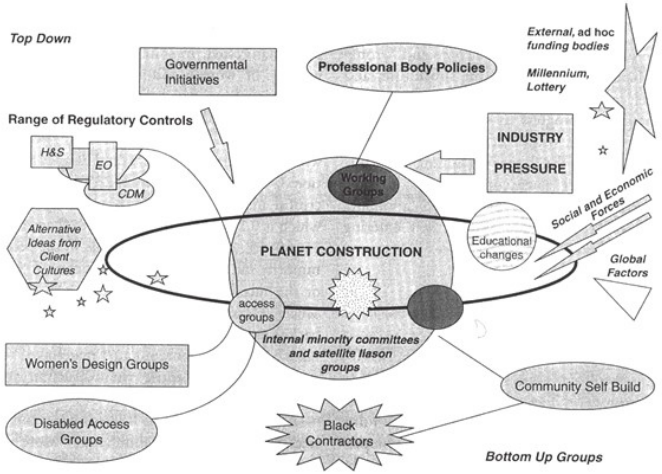


Figure 15.1 Diagram of Change Agents Impinging on Planet Construction.

Source: C. Greed (Introducing Planning, 2000)

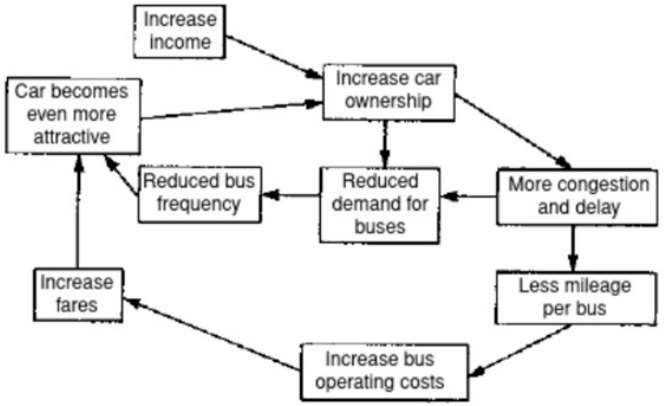


Figure 1.2 Car and public-transport vicious circle

Source: J.D. Ortúzar and J. Willumsen (Modelling Transportation 4th Ed, 2011)

Figure 1: Two Examples of Conceptual Models

How to use this note

The source for the document you are reading is an `R` Notebook. Notebooks are a form *literate programming*, a style of document that uses code to illustrate a discussion, as opposed to the traditional programming style that uses natural language to discuss/document the code. It flips around the usual technical writing approach to make it more intuitive and accessible.

Throughout the notes, you will find examples of code in segments of text called *chunks*. This is an example of a chunk, in the gray text box:

```
print("Hello, Discrete Choice Analysis!")
```

```
## [1] "Hello, Discrete Choice Analysis!"
```

If you are working with the Notebook version of the document, you can run the code by clicking the 'play' icon on the top right corner of the chunk. If you are reading the web-book version of the document, you will often see that the code has already been executed, right below the chunk. You can still try it by copying and pasting into your `R` or RStudio console. Whichever way you are working, you might want to give it a try now! You will see that the chunk of code above instructed `R` (and trough `R` the computer) to `print` (or display on the screen) some text.

Learning objectives

In this practice, you will learn about:

1. Different ways to measure stuff.
2. Importing data and packages in `R`.
3. Data classes, data types, and data transformations.
4. Indexing and data manipulation
5. Basic visualization.

Suggested readings

- Grolemund, G., Wickham, H. (2016) R for Data Science, **Chapters 3-5**, O'Reilly Media.
- Wickham, H. (2016) ggplot2: Elegant Graphics for Data Analysis, **Chapters 2-3**, Springer: New York.

Ways of measuring stuff

Previously we said that data are the raw material for modeling, but we did not say precisely what we meant by 'data'. You probably already have a working understanding of what 'data' means, but nonetheless we will begin with a definition. According to Merriam-Webster, data are:

> factual information (such as measurements or statistics) used as a basis for reasoning, discussion, or calculation

As an aside, it is interesting to note that Tukey's classic book Exploratory Data Analysis (Tuckey 1977) does not define 'data' in the glossary!

Measurement theory is a branch of mathematics concerned with observing and recording the *facts* about something. It is important to note that measurements are not the same as the thing being measured; however, we would like the measurements to be a reasonably close approximation of the thing being measured - with "reasonable" in this case meaning that reasonable people can agree that the measurements are a close approximation of the thing being measured (measurement theory breaks down in the face of unreasonable people). If this is not the case, the measurements obtained might be pretty useless, and an inadequate way to learn anything valuable from the thing we are measuring.

One fundamental contribution of the scientific method has been to produce standardized ways of measuring things. How would you measure the following things?

- The temperature at which water freezes.

- The temperature at which nitrogen freezes.
- The length of a trip.
- Blood donations.
- Different brands of peanut butter.
- The value of a two bedroom apartment.
- Someone's opinion regarding taxes.

Generally, there are multiple ways of measuring something, but not all of them are necessarily appropriate, partly because the scales of measurement may result in some loss of information. The interpretation of a measurement, as well, depends on what the scale is.

Two broad scales of measurement are as follows:

Categorical measurements

Categorical measurements assign a label or category to the thing being measured. For example, a way to measure different brands of peanut butter could be to measure their sugar content, their fat content, their consistency, and so on, and in this way describe what makes each brand unique. A different way to do this would be to label one brand "Spooky" and another "Peter's". This has the effect of reducing a large amount of information to a much simpler category. Is this loss of information inappropriate? Well, it really depends on what is the intended use of data! Categorical measurements are interesting because they may tell us something about the power of brands! They are also relevant when specific characteristics cannot be measured or observed.

Within the class of categorical variables there are two distinct scales of measurement:

- Nominal scale. When the categories do not follow a natural order. For example, there is no reason to say that the "Spooky" brand precedes "Peter's" or vice versa. Similarly, when measuring modes of travel "walking" is not intrinsically higher or lower or better or worse than "cycling" or "riding bus".

- Ordinal scale. When the categories follow a natural or logical sequence. A common way of measuring opinions is by means of the Likert scale, which classifies responses for instance as "strongly disagree", "disagree", "neutral", "agree", and "strongly agree". In this case, it is sensible to order the responses, since "strongly agree" is probably closer to "agree" than to "strongly disagree". Responses of this type are often represented by numbers, say, from 1 to 5, however, it is a mistake to treat the measurements as numbers instead of labels. When treated as numbers there is a temptation to think of the difference between 4 and 5 and the difference between 3 and 4 as being equivalent, when in fact the strength of disagreement could be stronger than the strength of agreement (or vice versa). In other words, the interval between "strongly disagree" and "disagree" may not be the same as "agree" and "strongly agree". With ordinal scales we do not know that, all that we know is that they measure a different opinion.

Sometimes, different measurement scales might represent different behavioral mechanisms, as Bhat and Pulugurta discuss in their comparison of categorical and ordinal measurements for vehicle ownership, which respectively correspond to unordered and ordered response mechanisms (Bhat and Pulugurta 1998).

Quantitative measurements

Quantitative measurements assign a number to an attribute, and the number quantifies the presence, strength, or intensity of the attribute. Within this class of variables, there are also two ways of measuring things.

- Interval scale. A quantity can be assigned to an attribute, the values follow an order, and their differences can be computed and remain constant. Temperature is typically measured in interval scale. The difference between 10 °C and 11 °C is the same as the difference between 25 °C and 26 °C. The intervals are meaningful. However, 0 °C does not imply the absence of temperature! Which is why measurements in Celsius and Fahrenheit do not coincide at zero. The lack of a natural zero for these scales means that the *ratios* between two values are not meaningful: 4 °C is not twice as hot as 2 °C, and −12 °C is not four times as cold as −3 °C.

- Ratio scale. When there is an absolute value of zero to the thing being measured (to indicate absence!), attributes can be measured in a ratio scale. This combines the features of the previous scales of measurement: a number is essentially a label that follows a logical order and with differences that are meaningful. In addition to that, the ratios of variables are meaningful. For example, twenty dollars are twice as valuable as ten, and zero is the absence of value. Weight is a way of measuring mass, and zero is the absence of mass. Two hundred kilograms is twice as much mass as one hundred kilograms.

It is important to understand the different scales of measurement to be able to choose the appropriate tools for each. More on this later. But first, we can bring some actual data to play with.

Importing data

There are several different ways of importing data in `R`. For this example, we will use a sample of a dataset that was analyzed by Whalen et al. (2013).

At the very beginning, it is good practice to clear the workspace, to ensure that there are no extraneous items there. The workspace is where objects reside in memory during a session with `R`. The function for removing variables from the workspace is `rm()`. Another useful function is `ls`, which retrieves a list of things in the workspace. So essentially we are asking `R` to remove all things in the workspace:

```
rm(list = ls())
```

Once that the workspace is empty, we can proceed to load a few packages that are useful. Packages are the basic units of reproducible code in the `R` multiverse. Packages allow a developer to create a self-contained unit of code that often is meant to achieve some task. For instance, there are packages in `R` that specialize in statistical techniques, such as cluster analysis, visualization, or data manipulation. Some packages can be miscellaneous tools, or contain mostly datasets. Packages are a very convenient way of maintaining code, data, and documentation, and also of sharing all these resources.

Packages can be obtained from different sources (including making them!). One of the reasons why `R` has become so successful is the relative facility with which packages can be distributed. A package frequently used is called `tidyverse`. The `tidyverse` is a collection of functions for data manipulation, analysis, and visualization. This package can be downloaded and installed in your personal library of `R` packages by using the function `install.packages`, as follows:

```
install.packages("tidyverse")
```

The function `install.packages` retrieves packages from the Comprehensive R Archive Network, or CRAN for short. CRAN is a collection of sites (accessible via the internet) that carry identical materials for distribution for `R`.

Installing a package is similar to acquiring a book for your library. The book is there, but if you want to use it, you need to bring it to your workspace, so to speak. The function for retrieving a package from the library is natural enough `library()`. For the moment, we need the following packages. If you have not done so, take a moment to install them, as illustrated in the previous chunk.

```
library(mlogit)
library(readr)
library(stargazer)
library(tidyverse)
```

Occasionally there are messages displayed when loading a package. These messages are informative (they ask you to cite them in a certain style) or may give you warnings, for instance that identically named functions exist in several packages.

The function that we need to read the sample dataset is `read_csv()`, which is part of the `tidyverse` package. Note that you can name the value (or output) of a function by using `<-`. In this case, we wish to read an external file, and assign the results to an object called `mc_mode_choice`:

```r
mc_mode_choice <- read_csv("Commute Mac.csv")
```

It is possible to quickly examine the contents of the object by means of the function `head()`, which prints the top few rows of the object, if appropriate. For example (showing only the first eight columns):

```r
head(mc_mode_choice[, 1:8])
```

```
## # A tibble: 6 x 8
##   RespondentID choice avcycle avwalk avhsr avcar timecycle timewalk
##          <dbl>  <dbl>   <dbl>  <dbl> <dbl> <dbl>     <dbl>    <dbl>
## 1    566872636      3       0      1     1     0      6.21     21.3
## 2    566873140      3       0      1     1     1      3.73     12.8
## 3    566874266      3       0      0     1     1 100000    100000
## 4    566874842      2       1      1     1     0      5.83     20
## 5    566881170      2       1      1     1     0      5.83     20
## 6    566907438      2       0      1     1     0 100000        10
```

Here we can see that what we just read is a table with several variables: an id, a variable called `choice`, some variables for time, etc. Hopefully, when reading data there is also a metadata file, a data dictionary or something that defines what the data are. For example, what does it mean for choice to be "3" or "1"? Was time measured in hours, seconds, minutes, or something else? These variables will be described below. Before that, however, we can use a different function to get further insights into the contents of the table by means of the `summary()` function (output not shown):

```r
summary(mc_mode_choice)
```

Or using packages such as `stargazer` to produce elegantly formatted tables (showing just the first five variables):

```r
stargazer(as.data.frame(mc_mode_choice[,1:8]), # stargazer works on data frames
          type = "latex", # the output is in latex
          header = FALSE, # do not print package version info in the output
          title = "Example of a table with summary statistics", # Title of table
          omit.summary.stat = c("N", "median"), # summary statistics to omit from output
          font.size = "small") # font size can be changed
```

Table 1: Example of a table with summary statistics

Statistic	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
RespondentID	570,566,454.000	3,786,118.000	566,872,636	567,814,188.0	574,925,212.0	587,675,235
choice	2.618	0.845	1	2	3	4
avcycle	0.275	0.447	0	0	1	1
avwalk	0.661	0.473	0	0	1	1
avhsr	0.961	0.194	0	1	1	1
avcar	0.547	0.498	0	0	1	1
timecycle	34,014.860	47,389.710	0.291	3.788	100,000.000	100,000.000
timewalk	37,364.730	48,384.330	1.000	13.665	100,000.000	100,000.000

This function will print a set of summary statistics for the variables in the table. The statistics presented by R are generally determined based on the scale of measurement of the variable - that is, the way the variable is coded. Presently, all summary statistics are calculated for quantitative variables (since all variables are coded that way as we will see below). We don't know if this makes sense, until we know what the variables are supposed to measure.

For example, the variable `choice` measures the use of one mode of transportation. There are four values in this scale: 1 through 4, with each indicating one of "Cycle", "Walk", "Car", or "HSR" (the local transit

agency in Hamilton, ON, Canada, where these data were collected). Check the results of the summary. What does it mean to say that the mean of choice is 2.618? Does this number make sense?

Data Classes in R

To understand why 2.618 of mode of transportation is not an appropriate summary measure for the variable `mode`, we need to know that `R` can work with different data classes, which include the following:

- Numerical
- Character
- Logical
- Factor

The ability to store information in different forms is important, because it allows `R` to distinguish what kind of operations are appropriate for a certain variable. Consider the following example (using indexing):

```
mc_mode_choice$choice[1] - mc_mode_choice$choice[4]
```

```
## [1] 1
```

Let us unpack what the chunk above did. First, we call our table `mc_mode_choice`. The string sign `$` is used to reference columns in the table. Therefore, we asked `R` to go and look up the column `choice` in the table `mc_mode_choice`. Finally, the number between square brackets `[]` asks `R` to retrieve a specific element out of the column, in this example the first element in that column and then the fourth element. This system of referring to elements in tables is called *indexing*. Most computer languages use it, but the syntax is different. Again: `$` refers to a column, and `[]` is used to call elements in that column.

As we can see, the difference between the two values (elements) retrieved is 1. But what is the meaning of "cycle" minus "walk", for instance?

In reality, the variable `choice` was measured as a nominal variable: it just corresponds to a label indicating what mode was chosen by a respondent. But `R` does not know this. Before `R` can treat it as a nominal variable, the numbers need to be converted to a *factor*. Factors are the way `R` stores categorical variables (both nominal and ordinal). To convert the variable `choice` to a factor, we use the `factor()` function:

```
mc_mode_choice$choice <- factor(mc_mode_choice$choice, labels = c("Cycle", "Walk", "HSR", "Car"))
```

In the chunk above, we ask `R` to replace the contents of `mc_mode_choice$choice` with the value (output) of the function `factor`. Factor takes the contents of `mc_mode_choice$choice` and converts to a factor with labels as indicated by the argument `labels =` (the function `c()` is used to concatenate several values).

We can summarize the result, by using the `summary()` function but only for this variable:

```
summary(mc_mode_choice$choice)
```

```
## Cycle  Walk   HSR   Car
##    48   711   336   281
```

Now the summary is appropriate categorical variable, and is a table of frequencies: as seen there, there were 48 respondents who chose "Cycle", 711 who chose "Walk", and so on. What if we tried to calculate the difference?

```
mc_mode_choice$choice[1] - mc_mode_choice$choice[4]
```

```
## Warning in Ops.factor(mc_mode_choice$choice[1], mc_mode_choice$choice[4]): '-'
## not meaningful for factors
```

```
## [1] NA
```

The message indicates that the operation we tried to perform is not meaningful for factors. As long as `R` knows the appropriate measurement scale for your variables, it will try to steer you away from doing silly things with them.

Other variables included in this table relate to time. These variables measure the duration in minutes (actual or imputed) for trips by different modes. For example, `timecycle` is the duration of a trip by bicycle for the journey reported by the respondent.

Let us summarize this variable again:

```
summary(mc_mode_choice$timecycle)
```

```
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max.
##      0.29      3.79      5.83  34014.86 100000.00 100000.00
```

Notice that the shortest trip by bicycle would be less than a minute long, whereas the maximum is $100,000$ minutes long. Wait, what? That is over $1,600$ hours long. Is that even possible? In fact, no, it is not. The reason for these values is that when the original data were coded, whenever a respondent said that cycling was not a mode that was available to them, the time was coded as a very large and distinctive value. There were no trips taking $100,000$ minutes, this is just a code for "information not available". One problem with this manner of coding is that `R` does not know that the information is actually missing, but rather thinks it is a legitimate quantity. As a consequence, the mean is tens of thousands of minutes, despite the fact that half of all trips by bicycle were measured at less than 6 minutes long (see the median).

Next we will see a way to address this. One last thing before doing so: you can check the class an object with the function `class`

```
class(mc_mode_choice$choice)
```

```
## [1] "factor"
```

```
class(mc_mode_choice$timecycle)
```

```
## [1] "numeric"
```

The RStudio also allows you to quickly see the variables and their format in the environment panel.

More on indexing and data manipulation

Indexing is a way of making reference to elements in a data object. There are numerous indexing methods in `R` that are appropriate for specific objects. Tables such as `mc_mode_choice` (called *data frames* in R) can be indexed in a few different ways. For example the next three chunks are equivalent in that they call the second column (`choice`) and in that column the second element:

```
mc_mode_choice[2, 2]
```

```
## # A tibble: 1 x 1
##   choice
##   <fct>
## 1 HSR
```

```
mc_mode_choice$choice[2]
```

```
## [1] HSR
## Levels: Cycle Walk HSR Car
```

```
mc_mode_choice[["choice"]][2]
```

```
## [1] HSR
## Levels: Cycle Walk HSR Car
```

It is also possible to index by ranges of values. For example, the next chunks retrieves rows 2 to 5 from columns 7 and 8:

```
mc_mode_choice[2:5, 7:8]
```

```
## # A tibble: 4 x 2
##   timecycle timewalk
##       <dbl>    <dbl>
## 1      3.73     12.8
## 2 100000    100000
## 3      5.83     20
## 4      5.83     20
```

Indexing is useful to subset data selectively. For example, we know that travel times coded as 100,000 are actually cases where the corresponding mode was not available. Suppose that we wanted to summarize travel time by bicycle but without those cases. We can use logical statements when indexing. We could tell `R` to retrieve only those values that meet a certain condition. In the next chunk, we save the results of this to a new variable:

```
time.Cycle.clean <- mc_mode_choice$timecycle[mc_mode_choice$timecycle != 100000]
```

where `!=` is `R` for *not equal to*. In other words, "find all values *not equal to* 100000, and retrieve them".

The result of this is a numeric object:

```
class(time.Cycle.clean)
```

```
## [1] "numeric"
```

If we summarize this object now:

```
summary(time.Cycle.clean)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.2914  2.9141  4.3711  4.8957  5.8282 45.0000
```

The summary statistics are much more sensible: the longest trip by bicycle was measured at 45 minutes, and the mean trip at less than 5 minutes.

Indexing is a powerful technique, but can be cumbersome (`mc_mode_choice$timecycle[mc_mode_choice$timecycle != 100000]`!). The package `dplyer` (part of the `tidyverse`) provides a grammar for data manipulation that is more intuitive. We will explore three of its elements here, namely the pipe operator (`%>%`), `select`, and `filter`.

Suppose that we wanted to select two of the time variables, for cycling and walking, and wanted to retrieve only values other than the offending 100,000, and save these values in a new object called `time.Active.clean`. In the grammar of `dplyr`, this is done as follows:

```
time.Active.clean <- mc_mode_choice %>%
  select(c("timecycle", "timewalk")) %>%
  filter(timecycle != 100000 & timewalk != 100000)
```

In natural language this would be something like "take mc_mode_choice and select columns timecycle and timewalk; pass the result to filter and retrieve all rows that meet the conditions timecycle != 100000 AND timewalk != 100000". The verb `select` is used to select columns from a data frame, and the verb `filter` to filter rows.

The alternative, using indexing would look something like this:

```
time.Active.clean.the.hard.way <- mc_mode_choice[mc_mode_choice$timecycle != 100000 &
                                                  mc_mode_choice$timewalk != 100000, 7:8]
```

The expression becomes more convoluted and not as easy to read. It is also easier to make mistakes when writing it.

Compare the summaries of the two data frames, to make sure that they are identical:

```r
summary(time.Active.clean)
```

```
##    timecycle          timewalk
##  Min.   : 0.2914   Min.   : 1.00
##  1st Qu.: 2.9141   1st Qu.:10.00
##  Median : 4.3711   Median :15.00
##  Mean   : 4.5852   Mean   :16.10
##  3rd Qu.: 5.8282   3rd Qu.:20.00
##  Max.   :17.4845   Max.   :62.11
```

```r
summary(time.Active.clean.the.hard.way)
```

```
##    timecycle          timewalk
##  Min.   : 0.2914   Min.   : 1.00
##  1st Qu.: 2.9141   1st Qu.:10.00
##  Median : 4.3711   Median :15.00
##  Mean   : 4.5852   Mean   :16.10
##  3rd Qu.: 5.8282   3rd Qu.:20.00
##  Max.   :17.4845   Max.   :62.11
```

The grammar of data manipulation in `dplyr` is a powerful way of working with data in an intuitive way. We will find other aspects of this, but for the time being you are welcome to consult more about `dplyr` here

Visualization

The last item in this section is related to data visualization. Humans are very much visual creatures, and much can be learned from *seeing* the data. For example, the data frame, in essence a table, is informative in many ways, but not particularly conducive to observe trends or regularities in the data. The summary statistics are also informative, but partial, and do not convey information to the same effect as a statistical plot.

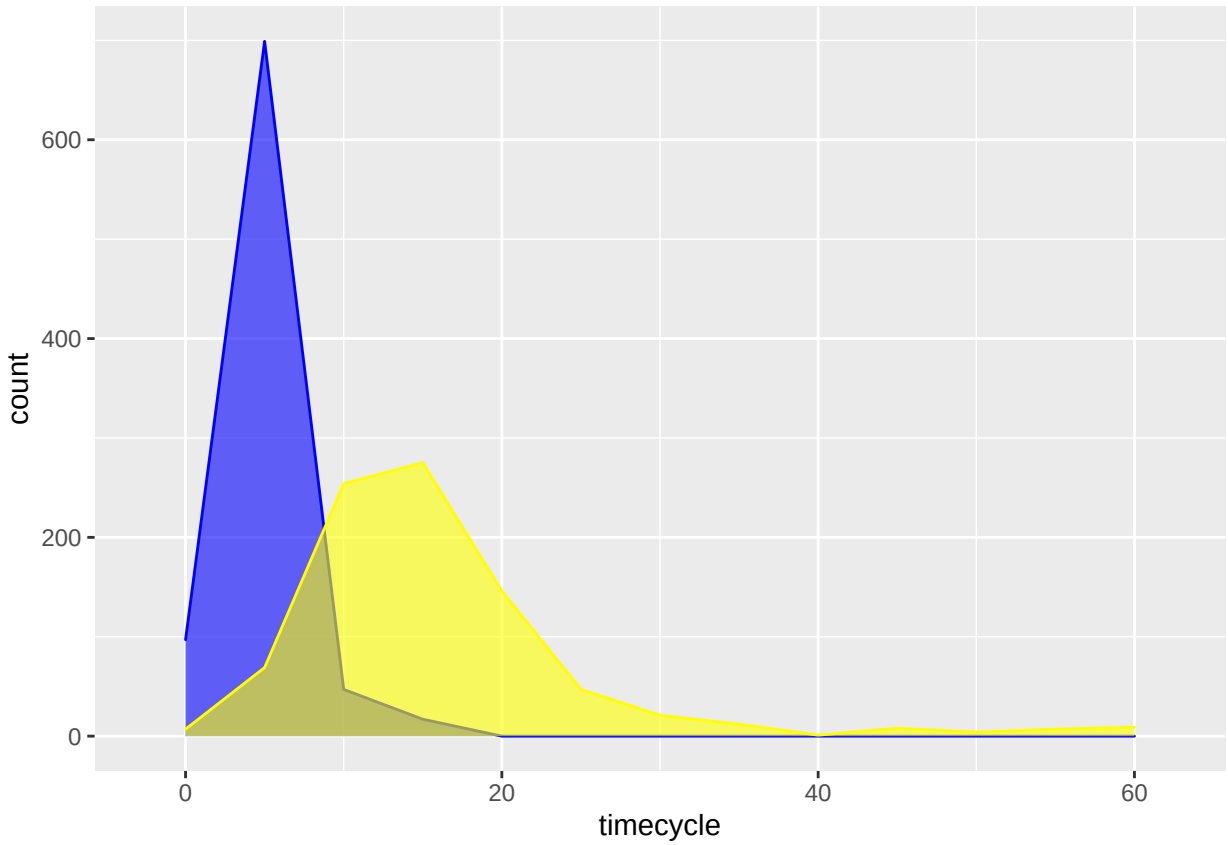
Take the following list of summary statistics:

```r
summary(time.Active.clean)
```

```
##    timecycle          timewalk
##  Min.   : 0.2914   Min.   : 1.00
##  1st Qu.: 2.9141   1st Qu.:10.00
##  Median : 4.3711   Median :15.00
##  Mean   : 4.5852   Mean   :16.10
##  3rd Qu.: 5.8282   3rd Qu.:20.00
##  Max.   :17.4845   Max.   :62.11
```

Now, compare to the following plot:

```r
ggplot(data = time.Active.clean) + geom_area(aes(x = timecycle), stat = "bin", binwidth = 5,
    fill = "blue", color = "blue", alpha = 0.6) + geom_area(aes(x = timewalk), stat = "bin",
    binwidth = 5, fill = "yellow", color = "yellow", alpha = 0.6)
```



The plot above was created using a package called `ggplot2`, also part of `tidyverse`. This package implements a grammar of graphics, and offers a very flexible way of creating plots in `R`.

`ggplot2` works by layering a series of objects, beginning with a blank plot, to which we can add things. The command to create a plot is `ggplot()`. This command accepts different arguments. For instance, we can pass data to it in the form of a data frame. We can also indicate different *aesthetic* values, that is, the things that we wish to plot. None of this is plotted, though, until we indicate which kind of *geom* or geometric object we wish to plot. For instance, you can see above that to create the figure we used `geom_area`. The corresponding geometric object is essentially a smoothed histogram.

Let us break down these instructions.

First we ask `ggplot2` to create a plot that will use the data frame `time.Active.clean`. We will name this object `p`:

```
p <- ggplot(data = time.Active.clean)
```

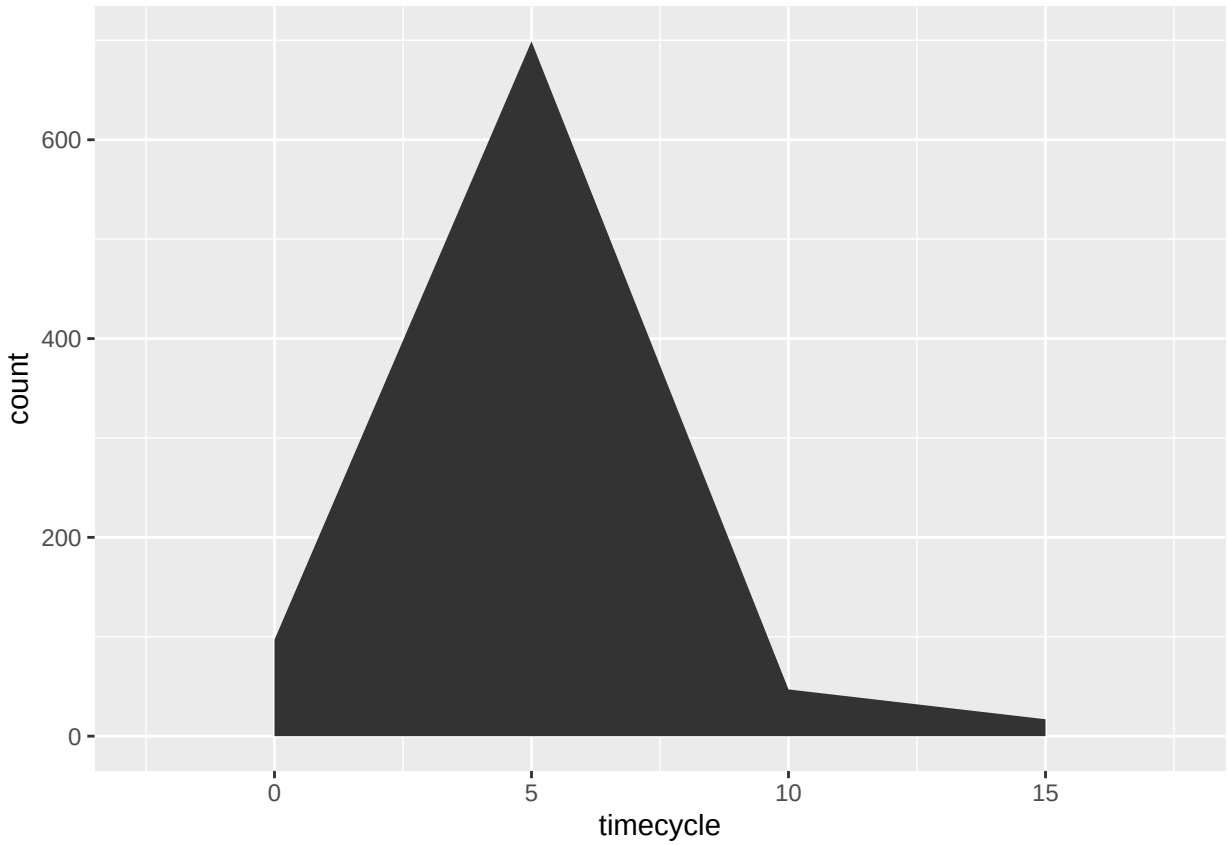
Notice how `ggplot2` creates a blank plot, but it has yet to actually render any of the population information in there:

```
p
```

We have yet to tell `ggplot2` what the x axis is, what the y axis is, what should be plotted in the x axis, and so on.

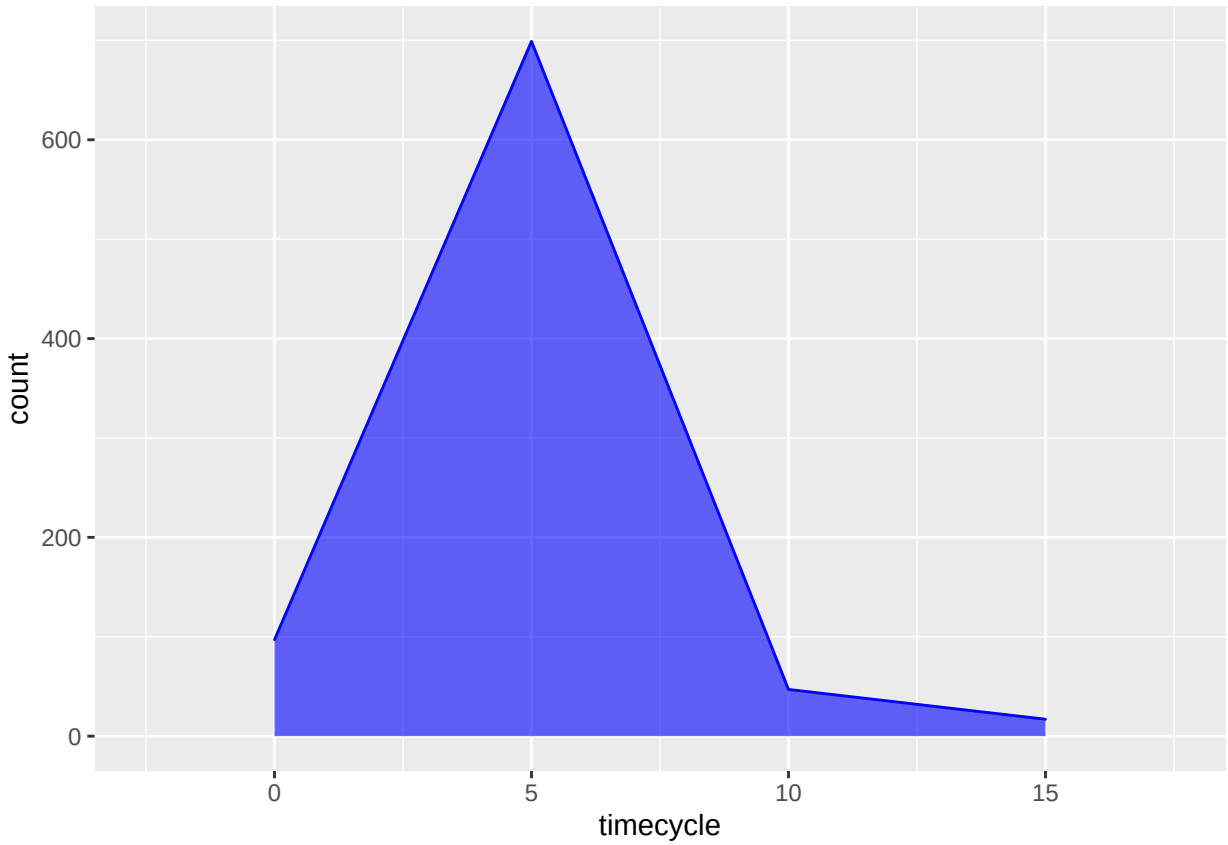
We layer elements on a plot by using the `+` sign. It is only when we tell the package to add some geometric element that it renders something on the plot. In the previous case, we told `ggplot2` to use the `geom_area` to create a smoothed histogram. Next, we need to indicate which `aes` (short for *aesthetics*) we wish to plot. The aesthetics map aspect of the dataset to the plot. For instance, by saying that the aesthetics include something for x, we tell `ggplot2` that that something should be mapped to the x axis (in this case one of the time variables in the data frame). The second argument is `stat = 'bin'`, which indicates that there is a statistical operation that happens, namely the data are binned for smoothing (small bins lead to less smoothing, large bins lead to more smoothing; try it!). After playing around with a few bin values, a width of 5 was selected:

```
p +
  geom_area(aes(x = timecycle),
            stat = "bin",
            binwidth = 5)
```



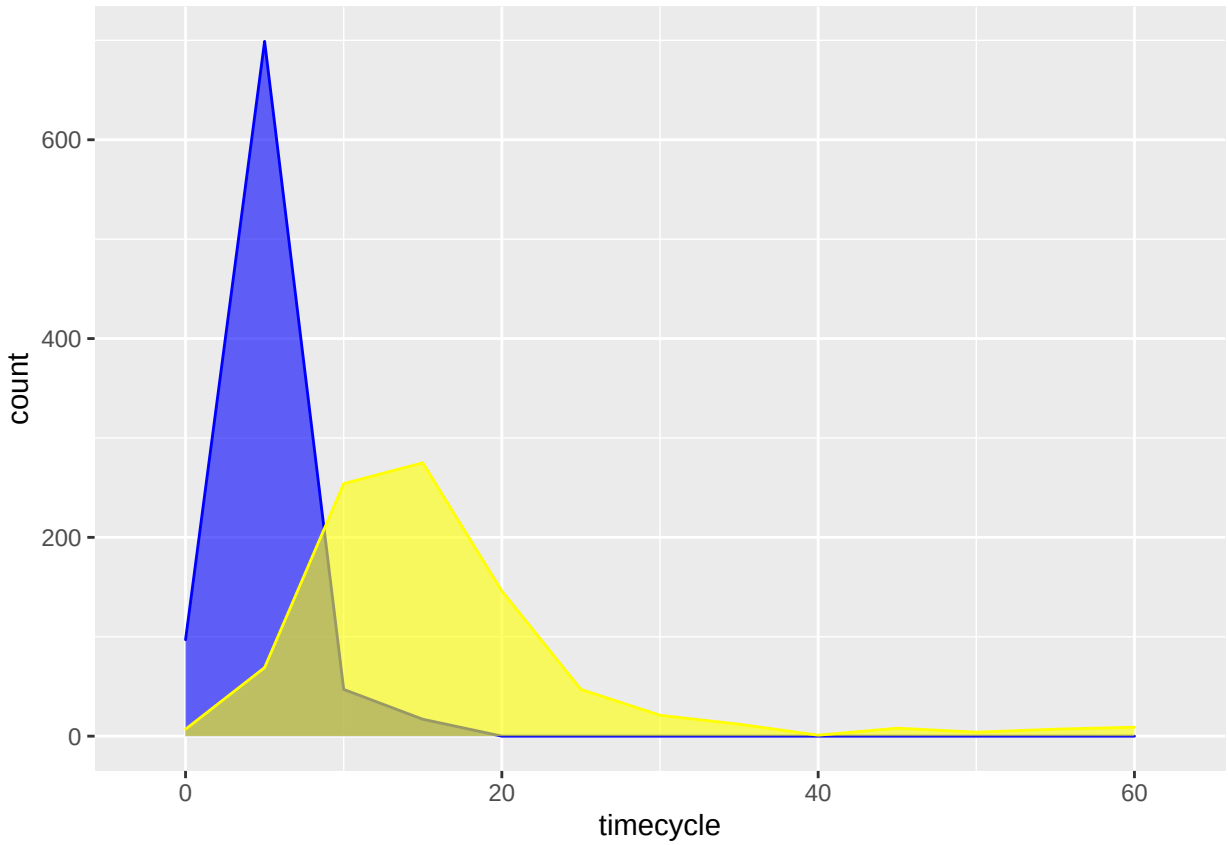
To improve the appearance of the plot, we also asked that the geom be rendered using a named color (blue for the color of the line, and also blue for the fill), and that it be transparent (the argument `alpha` controls opacity; a value of zero is transparent, a value of 1 is solid):

```
p +
  geom_area(aes(x = timecycle),
            stat = "bin",
            binwidth = 5,
            fill = "blue",
            color = "blue",
            alpha = 0.6)
```



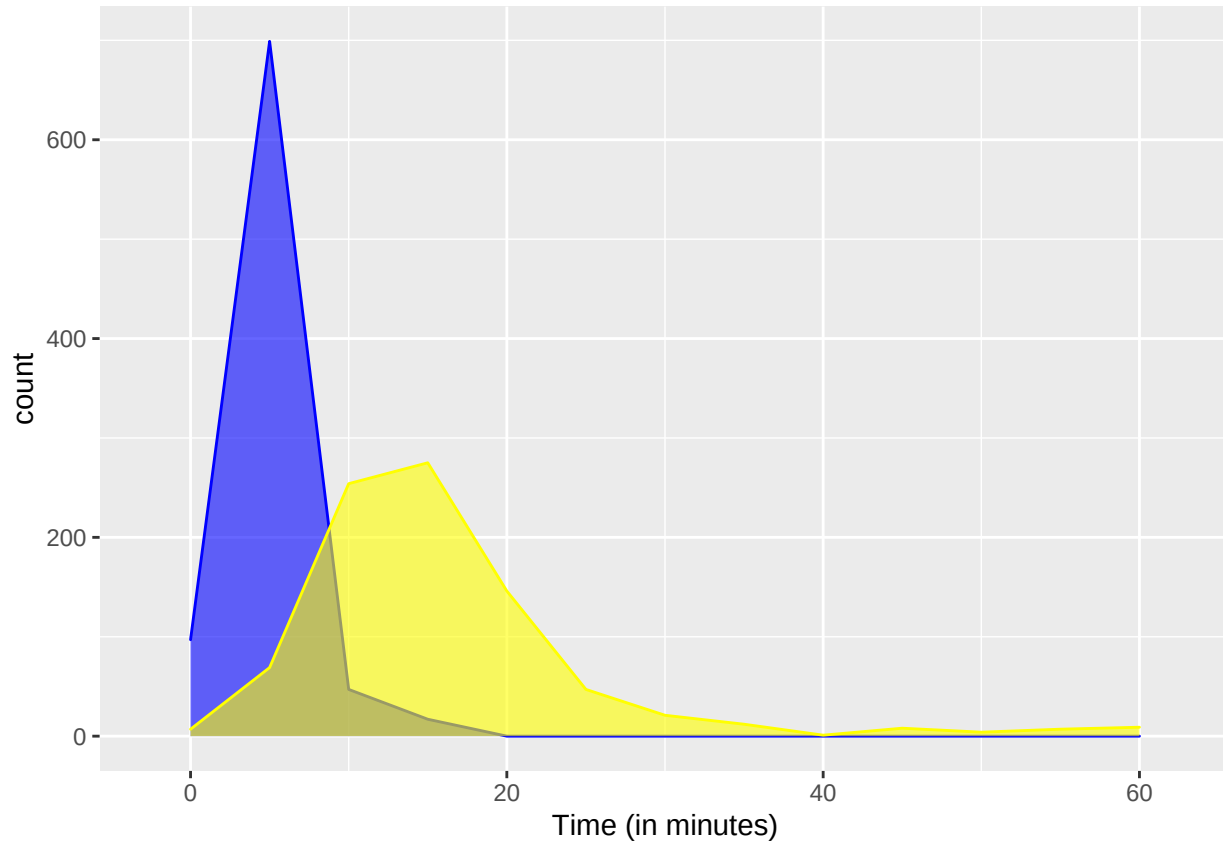
The final plot was obtained by layering a second geom (in yellow) for a different variable (time by walking), so that we could compare them:

```
ggplot(data = time.Active.clean) +
  geom_area(aes(x = timecycle),
            stat = "bin",
            binwidth = 5,
            fill = "blue",
            color = "blue",
            alpha = 0.6) +
  geom_area(aes(x = timewalk),
            stat = "bin",
            binwidth = 5,
            fill = "yellow",
            color = "yellow",
            alpha = 0.6)
```



Notice that the x axis is labeled as "timecycle" despite the fact that the plot also includes time by walking. This can be fixed by changing the label as follows:

```
ggplot(data = time.Active.clean) +
  geom_area(aes(x = timecycle),
            stat = "bin",
            binwidth = 5,
            fill = "blue",
            color = "blue",
            alpha = 0.6) +
  geom_area(aes(x = timewalk),
            stat = "bin",
            binwidth = 5,
            fill = "yellow",
            color = "yellow",
            alpha = 0.6) +
  xlab("Time (in minutes)")
```



What do we learn from this plot? Would it have been possible to learn the same from the summary statistics? Which was more effective, the plot or the summary statistics?

The plot above is an example of a *univariate* plot, since it is created to display the distribution of a single variable, not the way two or more variables relate. Imagine now that you would like to see how mode choice and sidewalk density at the place of residence relate. An appropriate statistical plot for two variables, one of which is nominal (`choice`) and another that is continuous (`side_den`), is the boxplot.

Before creating the plot, let us summarize these two variables (notice the use of the pipe operator):

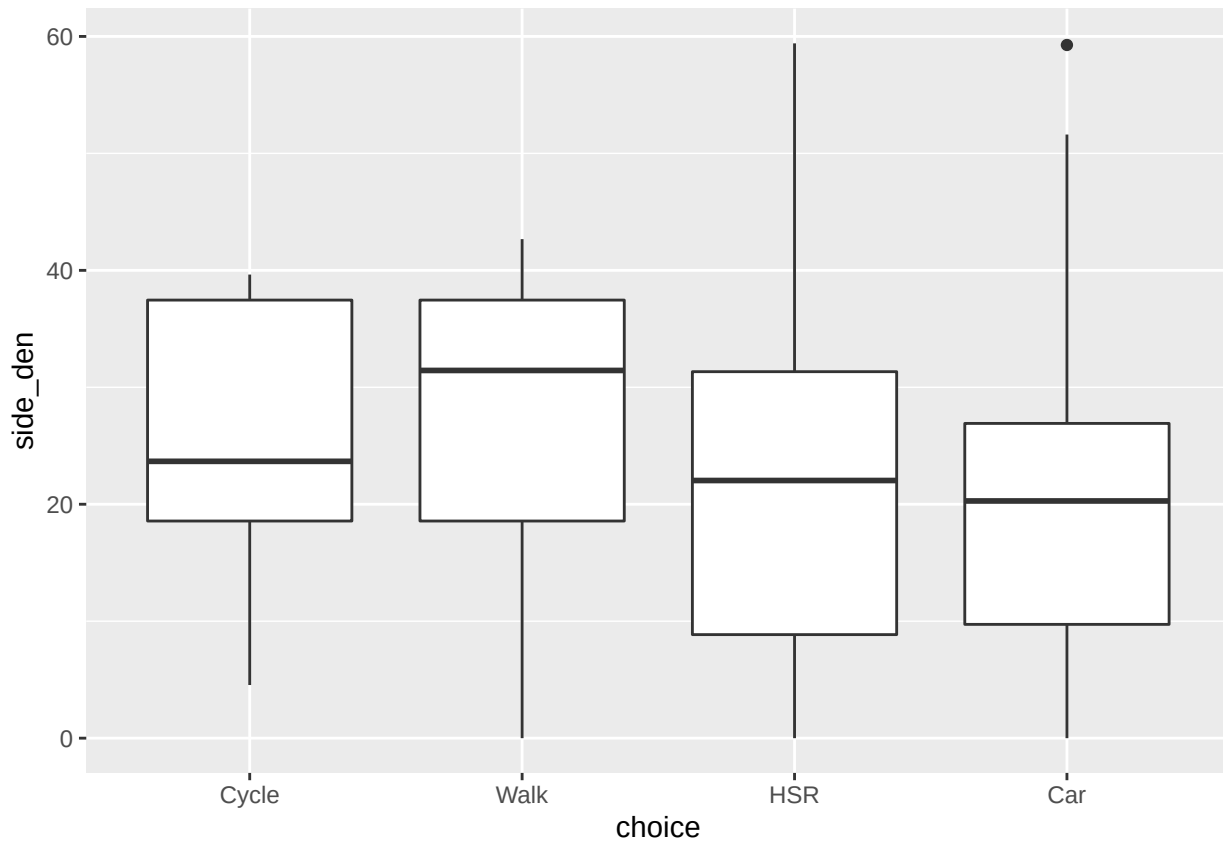
```
mc_mode_choice %>%
  select(c("choice", "side_den")) %>%
  summary()
```

```
##    choice       side_den
##  Cycle: 48   Min.   : 0.00
##  Walk :711   1st Qu.:18.19
##  HSR  :336   Median :22.63
##  Car  :281   Mean   :24.18
##              3rd Qu.:35.70
##              Max.   :59.41
```

Sidewalk density is measured in km/km^2.

We will create the boxplot next. We begin by defining a `ggplot2` object with the data frame and aesthetics that we wish to use. In this case, we want to plot the categorical variable in the x axis and the quantitative variable in the y axis:

```
ggplot(data = mc_mode_choice,
       aes(x = choice,
           y = side_den)) +
  geom_boxplot()
```



What do we learn from this plot? Could we have derived a similar insight from the summary statistics?

There are many different geoms that can be used in `ggplot2`. You can always consult the help/tutorial files by typing `??ggplot2` in the console. See:

```
?ggplot2
```

You can also check the `ggplot2` Cheat Sheet for more information on how to use this package.

A last note. Many other visualization alternatives (for instance, Excel) provide point-and-click functions for creating plots. In contrast, creating plots programmatically, as we illustrated in this chapter with `ggplot2` in `R`, requires that the plot be created by meticulously instructing the package what to do. While this is more laborious, it also means that you have complete control over the creation of plots, which in turn allows you to create more flexible and creative visuals. You can also easily recreate the figures if your data changes, for instance if you are analyzing the same data set after cleaning an error in your initial database, or a different data set with similar variables.

Exercise

Questions

1. Define "model".
2. Why are models on a 1-to-1 scale undesirable?

3. Invoke dataset `Mode` from package `mlogit`. To do this you need to first load the package. This is a dataset with choices about mode of transportation.

Once you have loaded the dataset, answer the following questions:

4. Describe this dataset. How many variables are there and of which type (i.e., categorical/quantitative)?
5. How many different modes of transportation are in this data set? What is the most popular mode? What is the least popular mode?
6. In general, what is the most expensive mode? The least expensive?
7. Create a plot showing the univariate distributions of time by car and time by bus. Discuss.
8. How do choices relate to cost by the different modes?

References

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Whalen, Kate E., Antonio Páez, and Juan A. Carrasco. 2013. "Mode Choice of University Students Commuting to School and the Role of Active Travel." Journal Article. *Journal of Transport Geography* 31: 132–42. https://doi.org/http://dx.doi.org/10.1016/j.jtrangeo.2013.06.008.

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